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Assignment 1

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Quiz questions

1. By convention what is the charge of glass and amber?
2. Why do conductors conduct electricity better than insulators?
3. The electric field between two charged plates is $E = \sigma/(2\epsilon_0)$. True or false?
4. Which of the following statements are correct regarding Gauss' Law (Select all that apply)
   (a) Gauss's Law states that the net electric flux through any closed surface is equal to the enclosed electric charge divided by the permittivity of the medium.
   (b) Gauss's Law applies only to isolated point charges.
   (c) Gauss's Law can be used to determine the electric field produced by continuous charge distributions, such as charged wires or planes.
   (d) Gauss's Law is a consequence of Coulomb's Law.
   (e) Gauss's Law holds true for any shape of the closed surface.
   (f) Gauss's Law is applicable to situations involving both electric and magnetic fields.
5. What are the three types of electrostatic charging?

Quiz answers

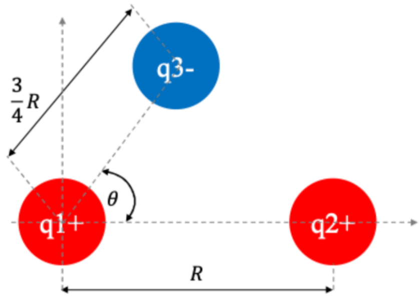
1. Following Benjamin Franklins convention, glass becomes positively charged and amber becomes negatively charged, when charged. This charge results from the triboelectric effect and is therefore not present before rubbing the materials.
2. Conductors conduct electricity as they have mobile electrons (i.e. no real band gap for metals). For insulators the opposite is the case – these have no mobile electrons and can therefore not carry a current (i.e. a large band gap between the valence and conduction bands)
3. False, the electric field between two plates with charge density σ is $E = \sigma/\epsilon_0$. $E = \sigma/(2\epsilon_0)$ is the formula for the electric field due to an infinite sheet.
4. (a) False, Gauss' law states that net electric flux through any closed surface equals the enclosed charge divided by the permittivity of free space ϵ_0.
   (b) False, Gauss' law applies for all charge configurations
   (c) True
   (d) True,
   $$\Phi = \frac{q}{\epsilon_0}$$
   can be directly calculated using the flux integral (assuming rotational symmetry about the middle of ones conductor)
   $$\Phi = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A}$$
   and Coulomb's law
   $$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}.$$
   (e) True, as long as the surface is closed.
   (f) False, Gauss' law only concerns itself with electric fields. No contribution from the magnetic field is present anywhere. For magnetism Gauss' law sounds
   $$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A} = 0$$

i.e. the magnetic flux through a closed surface is zero (law of no magnetic monopoles).

5. The three types are:
   (a) Charging by friction (Triboelectric effect)
   (b) Charging by contact
   (c) Charging by induction

Problem 1A

Consider $q_1 = 1{,}6 \cdot 10^{-19}\,\mathrm{C}$, $q_2 = 3{,}2 \cdot 10^{-19}\,\mathrm{C}$, and $q_3 = -3{,}2 \cdot 10^{-19}\,\mathrm{C}$ with distance $R = 0{,}02\,\mathrm{m}$ and $\theta = 60°$. What is the net electrostatic force on particle 1 from particles 2 and 3?



Givens

$q_1 = 1{,}6 \cdot 10^{-19}\,\mathrm{C}$	Charge of particle 1
$q_2 = 3{,}2 \cdot 10^{-19}\,\mathrm{C}$	Charge of particle 2
$q_3 = -3{,}2 \cdot 10^{-19}\,\mathrm{C}$	Charge of particle 3
$R = 0{,}02\,\mathrm{m}$	Distance between particles 1 and 2
$\theta = 60°$	Angle between charges 2 and 3 measured from charge 1

Derivation

Coulomb's law states that the electrostatic force $\boldsymbol{F}_{12}$ between two point charges q_1 and q_2 separated by $\boldsymbol{r}_{12}$ is

$$\boldsymbol{F}_{12} = k\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}.$$

We first find the unit vectors $\hat{r}_{12}$ and $\hat{r}_{13}$ as

$$\begin{aligned}
\hat{r}_{12} &= -1\hat{\imath} \\
\hat{r}_{13} &= -\cos 60°\,\hat{\imath} - \sin 60°\,\hat{\jmath} = -0{,}5\hat{\imath} - 0{,}866\hat{\jmath}.
\end{aligned}$$

This means that the force on particle 1 from particle 2 and 3 is:

$$\begin{aligned}
\boldsymbol{F}_{12} &= 8{,}99 \cdot 10^{9}\,\mathrm{N\,m^2/C^2} \cdot \frac{1{,}6 \cdot 10^{-19}\,\mathrm{C} \cdot 3{,}2 \cdot 10^{-19}\,\mathrm{C}}{(0{,}2\,\mathrm{m})^2} \cdot (-\hat{i}) \\
&= -1{,}15 \cdot 10^{-24}\,\mathrm{N}\hat{\imath} \\
\boldsymbol{F}_{13} &= 8{,}99 \cdot 10^{9}\,\mathrm{N\,m^2/C^2} \cdot \frac{1{,}6 \cdot 10^{-19}\,\mathrm{C} \cdot \left(-3{,}2 \cdot 10^{-19}\,\mathrm{C}\right)}{(0{,}015\,\mathrm{m})^2} \cdot \left(-0{,}5\hat{\imath} - 0{,}866\hat{\jmath}\right) \\
&= -2{,}05 \cdot 10^{-24}\,\mathrm{N}\left(-0{,}5\hat{\imath} - 0{,}866\hat{\jmath}\right) \\
&= 1{,}02 \cdot 10^{-24}\,\mathrm{N}\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\mathrm{N}\hat{\jmath}.
\end{aligned}$$

Using the principle of superposition the total force on particle 1 is then:

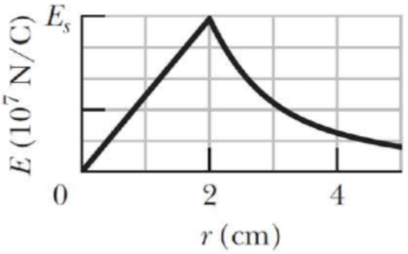
$$\boldsymbol{F}_{1,\mathrm{net}} = \boldsymbol{F}_{12} + \boldsymbol{F}_{13} = \left(1{,}02 \cdot 10^{-24}\,\mathrm{N} - 1{,}15 \cdot 10^{-24}\,\mathrm{N}\right)\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\mathrm{N}\hat{\jmath} = -0{,}13 \cdot 10^{-24}\,\mathrm{N}\,\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\mathrm{N}\,\hat{\jmath}.$$

$$\boxed{\boldsymbol{F}_{1,\text{net}} = -0{,}13 \cdot 10^{-24}\,\mathrm{N}\,\hat{\imath} + 1{,}77 \cdot 10^{-24}\,\mathrm{N}\,\hat{\jmath}}$$ RESULT

Problem 2A

The figure on the right, gives the magnitude of the electric field inside and outside a sphere with a positive charge distributed uniformly throughout its volume. The scale of the vertical axis is set by $E_s = 10 \cdot 10^7 \, \frac{\text{N}}{\text{C}}$.

1. What is the charge on the sphere?
2. What is the field magnitude at $r = 8{,}0\,\text{m}$?



Givens

$$\begin{array}{r|l} E_s = 10 \cdot 10^7 \, \frac{\text{N}}{\text{C}} & \text{Scale of vertical axis} \\ R = 2\,\text{cm} & \text{Radius of sphere} \end{array}$$

Derivation

As charge exists in the sphere the sphere itself must be insulating (no charge exists within a conducting sphere) from the figure we see that the radius of the sphere is 2 cm . Just on the edge (or outside) of a uniformly charged sphere, the field will be as if the sphere was a point charge:

$$E_s = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}.$$

Solving for the charge Q we obtain

$$Q_s = 4\pi\epsilon_0 R^2 E_s = 4\pi \cdot 8{,}85 \cdot 10^{-12}\,\text{C}^2/\text{Nm}^2 \cdot (2\,\text{cm})^2 \cdot 10 \cdot 10^7 \, \frac{\text{N}}{\text{C}} = 4{,}4 \cdot 10^{-7}\,\text{C}.$$

At $r = 8{,}0\,\text{m}$ we're outside of the sphere, where the E-field is:

$$E(8{,}0\,\text{m}) = \frac{1}{4\pi\epsilon_0} \frac{Q_s}{r^2}.$$

Plugging in all the known values we get:

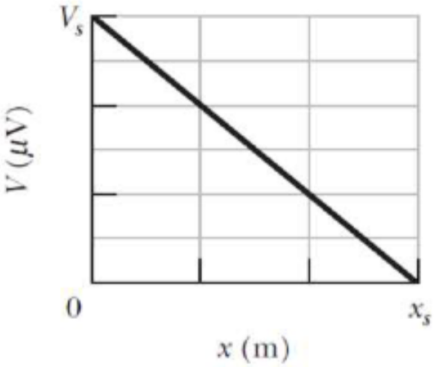
$$E(8{,}0\,\text{m}) = \frac{1}{4\pi \cdot 8{,}85 \cdot 10^{-12}\,\text{C}^2/\text{Nm}^2} \cdot \frac{4{,}4 \cdot 10^{-7}\,\text{C}}{(8{,}0\,\text{m})^2} = 61{,}8 \, \frac{\text{N}}{\text{C}}.$$

$$\boxed{\begin{aligned} Q_s &= 4{,}4 \cdot 10^{-7}\,\text{C} \\ E(8{,}0\,\text{m}) &= 61{,}8 \, \frac{\text{N}}{\text{C}}. \end{aligned}}$$

RESULT

Problem 3A

The figure on the right gives the electric potential $V(x)$ along a copper wire carrying a uniform current, from a point of higher potential, $V_s = 12\,\mu\text{V}$ at $x = 0$ to a point of zero potential at $x_s = 3\,\text{m}$. The wire has a radius of 2,20 mm. What is the current in the wire?
Hint: Use the resistivity of the copper $\rho_{\text{Cu}} = 1{,}68 \cdot 10^{-8}\,\Omega\,\text{m}$



Givens

$$
\begin{array}{r|l}
V_s = 12\,\mu\text{V} & \text{Potential at } x = 0 \\
x_s = 3\,\text{m} & \text{Distance for } V = 0 \\
r = 2{,}20\,\text{mm} & \text{Radius of wire} \\
\rho_{\text{cu}} = 1{,}68 \cdot 10^{-8}\,\Omega\,\text{m} & \text{Resistivity of copper}
\end{array}
$$

Derivation

Resistivity is defined as:

$$\rho = \frac{RA}{l}$$

where R is resistance, A is cross-sectional area and l is length. Solving this for the resistance R we obtain:

$$R = \frac{\rho_{\text{cu}} x_s}{\pi \cdot r^2}.$$

Plugging in known values we get:

$$R = \frac{1{,}68 \cdot 10^{-8}\,\Omega\,\text{m} \cdot 3\,\text{m}}{\pi \cdot (2{,}20\,\text{mm})^2} = 3{,}3 \cdot 10^{-3}\,\Omega.$$

Ohm's law states that the current is given by the ratio of the voltage difference and the resistance:

$$I = \frac{U}{R}.$$

Which gives:

$$I = \frac{V_s}{R} = \frac{12\,\mu\text{V}}{3{,}3 \cdot 10^{-3}\,\Omega} = 3{,}64\,\text{mA}.$$

$$\boxed{I = 3{,}64\,\text{mA}}$$ RESULT

Quiz questions

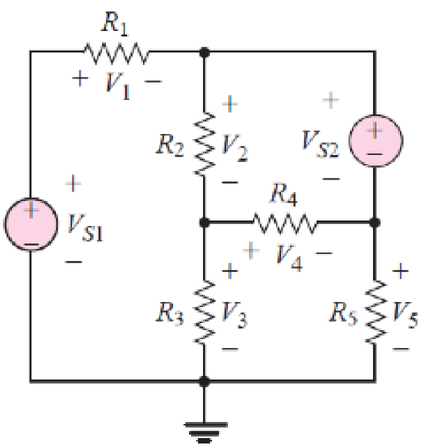
1. Resitivity is a temperature-independent quantity. True or False?
2. Kirchoff's current law states that the current flowing into a node (or a junction) must be equal to the current flowing out of it. True or False?
3. Kirchhoff's voltage law states that the sum of the voltage differences around any closed loop in a circuit must be zero. True or False?
4. The Earth is not an example of a reference voltage. True or False?
5. The ammeter must be placed in series with the element whose voltage is measured.

Quiz answers

1. False. Resistivity varies with temperature approximately according to $\rho - \rho_0 = \rho_0 \alpha (T - T_0)$ from a value ρ_0 at temperature T_0 to ρ at temperature T.
2. True.
3. True.
4. False, the earth is an example of a reference voltage (commonly called ground or zero volts).
5. True, an ammeter must always be connected in series as it has a very small internal resistance meaning placing it in parallel short-circuits the circuit.

Problem 1B

Use KVL to determine the unknown voltages V_1 and V_4 in the circuit. Known quantities in the circuit are $V_{S1} = 12\,\mathrm{V}$, $V_{S2} = -4\,\mathrm{V}$, $V_2 = 2\,\mathrm{V}$, $V_3 = 6\,\mathrm{V}$, and $V_5 = 12\,\mathrm{V}$.



Givens

$V_{S1} = 12\,\mathrm{V}$	Voltage supplied by source $S1$
$V_{S2} = -4\,\mathrm{V}$	Voltage supplied by source $S2$
$V_2 = 2\,\mathrm{V}$	Voltage drop across resistor $R2$
$V_3 = 6\,\mathrm{V}$	Voltage drop across resistor $R3$
$V_5 = 12\,\mathrm{V}$	Voltage drop across resistor $R5$

Derivation

Using KVL we observe the closed loop containing source $S1$ and resistors $R1, R2$, and $R3$. KVL states that the sum of the voltage differences around this loop must be zero. Hence,

$$V_{S1} - V_1 - V_2 - V_3 = 0 \implies V_1 = V_{S1} - V_2 - V_3.$$

So

$$V_1 = 12\,\mathrm{V} - 2\,\mathrm{V} - 6\,\mathrm{V} = 4\,\mathrm{V}.$$

We now consider the loop containing source $S2$ and resistors $R2$ and $R4$ and use KVL to get

$$V_{S2} - V_2 - V_4 = 0 \implies V_4 = V_{S2} - V_2.$$

So

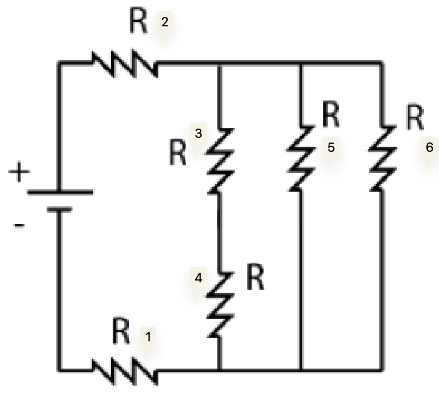
$$V_4 = -4\,\mathrm{V} - 2\,\mathrm{V} = -6\,\mathrm{V}.$$

$$\boxed{\begin{aligned} V_1 &= 4\,\mathrm{V} \\ V_4 &= -6\,\mathrm{V}. \end{aligned}}$$

RESULT

Problem 2B

Solve for the equivalent resistance R_{eq} of the whole circuit on the Figure on the right.



Derivation

For resistors in series the equivalent resistance is

$$R_{\text{eq}} = R_1 + R_2.$$

And for resistors in parallel their equivalent resistance is

$$R_{\text{eq}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}.$$

We start by considering the series resistors $R3$ and $R4$ and find their equivalent resistance as:

$$R_{34} = R_3 + R_4 = 2R.$$

For the three paralell resistors $R34$, $R5$ and $R6$ we obatin

$$\begin{aligned} R_{3456} &= \frac{1}{\frac{1}{R_{34}} + \frac{1}{R_5} + \frac{1}{R_6}} \\ &= \frac{1}{\frac{1}{2R} + 2 \cdot \frac{1}{R}} \\ &= \frac{1}{\frac{5}{2R}} \\ &= \frac{2}{5}R. \end{aligned}$$

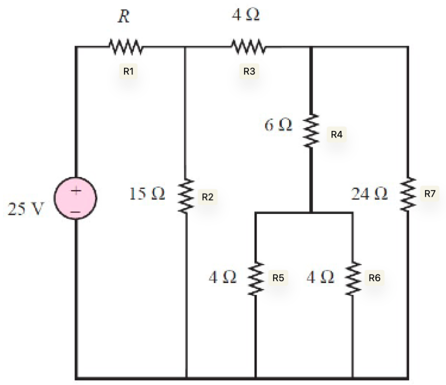
Now we are left with the three resistors $R1$, $R2$, and $R3456$ in series, which gives the total equivalent resistance:

$$R_{\text{eq}} = R_1 + R_2 + R_{3456} = 2R + \frac{2}{5}R = 2{,}4R.$$

$$\boxed{R_{\text{eq}} = 2{,}4R}$$ RESULT

Problem 3B

In the circuit on the figure the power absorbed by the $15\,\Omega$ resistor is $15\,\text{W}$. Find R.



Givens

$R_2 = 15\,\Omega$	Resistance of R2
$R_3 = 4\,\Omega$	Resistance of R3
$R_4 = 6\,\Omega$	Resistance of R4
$R_5 = 4\,\Omega$	Resistance of R5
$R_6 = 4\,\Omega$	Resistance of R6
$R_7 = 24\,\Omega$	Resistance of R7

Derivation

We use the formula for parallel resistors introduced in the last problem the equivalent resistance R_{56} of resistors $R5$ and $R6$ is

$$R_{56} = \frac{1}{\frac{1}{R_5} + \frac{1}{R_6}} = \frac{1}{\frac{1}{4\,\Omega} + \frac{1}{4\,\Omega}} = 2\,\Omega.$$

Which means the equivalent resistance R_{456} becomes:

$$R_{456} = R_4 + R_{56} = 2\,\Omega + 6\,\Omega = 8\,\Omega.$$

Now, we realize R7 is in parallel with the above equivalent resistance giving R_{4567} as

$$R_{4567} = \frac{1}{\frac{1}{R_{456}} + \frac{1}{R_7}} = \frac{1}{\frac{1}{8\,\Omega} + \frac{1}{24\,\Omega}} = 6\,\Omega.$$

$R3$ is in series with this so R_{34567} is

$$R_{34567} = R_3 + R_{4567} = 4\,\Omega + 6\,\Omega = 10\,\Omega.$$

Which is in parallel with $R2$ giving R_{234567} as:

$$R_{234567} = \frac{1}{\frac{1}{R_2} + \frac{1}{R_{34567}}} = \frac{1}{\frac{1}{10\,\Omega + \frac{1}{15\,\Omega}}} = 6\,\Omega.$$

So the total equivalent resistance of the circuit is:

$$R_{\text{eq}} = R_1 + R_{234567} = R_1 + 6\,\Omega.$$

The voltage drop across R2 can be found using the power law as:

$$P_2 = \frac{U_2^2}{R_2} \implies U_2 = \sqrt{P_2 R_2} = \sqrt{15\,\Omega \cdot 15\,\text{W}} = 15\,\text{V}.$$

Using KVL this means that the voltage drop across $R1$ is

$$V_s - V_1 - V_2 \implies V_1 = V_s - V_2 = 25\,\mathrm{V} - 15\,\mathrm{V} = 10\,\mathrm{V}.$$

This means the current over $R1$ is

$$I_1 = \frac{U_1}{R_1} = \frac{10\,\mathrm{V}}{R_1}.$$

The current supplied by the source is

$$I = \frac{U_s}{R_{\mathrm{eq}}} = \frac{25\,\mathrm{V}}{R_1 + 6\,\Omega}.$$

From KCL the current at the source must equal the current at $R1$, hence $I = I_1$ so

$$\begin{aligned}
\frac{10\,\mathrm{V}}{R_1} &= \frac{25\,\mathrm{V}}{R_1 + 6\,\Omega} \\
\frac{R_1}{10\,\mathrm{V}} &= \frac{R_1}{25\,\mathrm{V}} + \frac{6\,\Omega}{25\,\mathrm{C}} \\
\frac{3}{50\,\mathrm{V}} R_1 &= \frac{6\,\Omega}{25\,\mathrm{V}} \\
R_1 &= \frac{50\,\mathrm{V}}{3} \cdot \frac{6\,\Omega}{25\,\mathrm{V}} \\
R_1 &= 4\,\Omega.
\end{aligned}$$

$$\boxed{R_1 = 4\,\Omega}$$ RESULT